Changes and evidence for the second round revision

Working revision 1.27 • 3 October 2026 • Author verification required

This record compares the supplied version 1.26 manuscript and response with the replacement working revision. It records substantive changes and why they were necessary. The original .doc files remain unchanged. The new manuscript was reconstructed as an editable .docx from verified extracted text and a new section structure; its old layout and historical figures are not preserved as quantitative evidence.

Revision record

Original issue or claim	Replacement and evidence	Location
Imagined speech reconstruction title	Text-guided composition; no unknown utterance decoding evaluated	Title, Abstract, Section 1
Fractal-inspired or self-similar brain model	Removed unsupported geometric claim	Sections 1 and 2.7
LLM trained on autobiography	Retrieval and prompting; weights not fine-tuned	Section 2.2
Responses constrained to 20 words	Frozen responses exceed target; enforcement not claimed	Section 2.2, Table 1
Counter zero treated as task onset	Paired markers; 44/44 old blocks before first task	Section 2.1, events.csv, data_audit.csv
256 Hz from counter/block convention	Documented nominal 250 Hz and timestamp diagnostics	Section 2.1
Alphabetical ARPAbet-to-file mapping	Public-slide crosswalk; vowel/dialect approximations declared	Section 2.1, code/reanalyse.py
Files 39–43 as random gap sources	Those stimuli are NG R L W Y; use marked wait intervals	Section 2.3
One final phoneme per word presented as all phonemes	Primary all-phoneme design; final-only sensitivity	Sections 2.3 and 3.3
Condition-dependent contraction splitting	Same word token stream in both conditions	Section 2.3
Missing hand-transcribed probabilistic fixture	Frozen public CMUdict and declared BCIs/streetside overrides	inputs/pronunciations.json
Repeated sample and trial-indexing defects	Marker-based blocks and complete 125-sample windows	Sections 2.1 and 2.4
Overlapping Welch claim with 32-sample resolution	Explicit nonoverlapping 125-sample periodograms, 2 Hz bins	Section 2.4
Absolute CMRO2 treated as fractional change	$d = C/2.1 - 1$ and dimensional CBF conversion	Section 2.5
Gamma estimates measured oxygen metabolism	Dimensionless index primary; imposed physiological proxy	Sections 2.4–2.5
Derived eight outcomes as independent evidence	One deterministic driver; supporting outputs only	Sections 2.5 and 3.5
Channels n=16 or microgap draws as biological n	Conditional simulation variation, no population tests	Section 2.6
95% CI and exact MC p from treatment draws	Central simulation interval and MCSE; p values withdrawn	Abstract, Section 2.6, Tables 3–4
High rho interpreted as indistinguishable real metabolism	Rank similarity with nonzero error; proxy and reference limits	Sections 2.7 and 3.4
Reference includes reused construction blocks	Disjoint within-session trial roles 1/2/3–5	Section 2.3
Source operators vary with each recording	One common reference covariance and inverse	Section 2.7

Original issue or claim	Replacement and evidence	Location
Custom sphere described as validated MNE equivalent	Simplified radial sphere sensitivity; no anatomical validation	Section 2.7, Figure 3
Smoothness or biological pause effects inferred	Duration and window-order controls; effects concern composition	Sections 2.6 and 4
Old paired t tables and inconsistent Trial 2 values remain	All old inferential tables replaced from frozen results	Tables 2–5
Old GUI images treated as corrected evidence	Four new reproducible figures; old animations excluded	Figures 1–4
No live subjects used as sufficient ethics statement	Secondary human EEG provenance and approval items explicit	Declarations
Main URL treated as exact reproducible release	Stable approved release identifier still required	Data availability
Speculative scalability clinical or truly-random benefits	Removed; concrete independent validation requirements	Discussion and Conclusions

AI assistance in analysis-code development, manuscript revision, and figure-script preparation is disclosed in Section 2.9 and the figure legends. The disclosure still requires the authors to describe their own independent verification before submission.

Numerical traceability

Source mapping

Manuscript material	Machine-readable source
Abstract changes and Table 3	results.json summary where config=primary and variable=CMRO2; subtract 2.1 only for baseline index
Table 1	inputs/sentences.json
Table 2	data_audit.csv, events.csv, stimulus_labels.json, upstream_matches.json
Table 4	simulation_summary.csv where variable=CMRO2
Table 5	consistency.csv where kind=composite
Signal quality	signal_quality.csv
S1 constants	results.json constants and code/model.py
Figure 2	results.json summary
Figure 3	spectra.npz and source_maps.npz
Figure 4	public file DLR_0_1.txt and seed-0 assembly manifest

The mean CMRO2-proxy difference numerically equals the dimensionless index difference because the imposed scale has a slope of 1 μmol/(g·min) per index unit. This identity does not make those outputs independent or validate the physical scale. Raw CBF values in the CSV are mL/(g·min); the manuscript explains the ×100 conversion for the customary per-100-g unit.

Pending author decisions

Confirm the recording and montage metadata and the pronunciation crosswalk; review source-signal quality; establish participant/ethics and reuse facts; finalize declarations and contact details; approve the substantially narrowed study framing; and deposit the approved reproducible version. A marked-up journal manuscript must use the exact submitted source as its comparison base. The detailed changes above are not a claim that all of these external facts have been resolved.